

Sandeep Singh

Lead SRE | Kubernetes Platform Architect | GitOps, Cloud Infrastructure & Observability

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Lead Site Reliability Engineer with 10+ years of experience architecting and scaling Kubernetes platforms and cloud-native infrastructure across multi-region, **multi-cloud environments (AWS, GCP, Azure)**. Specialized in Infrastructure as Code (**Terraform**), **GitOps (ArgoCD)**, and automated CI/CD pipelines. Proven success in **reducing cloud costs by up to 30% through cluster autoscaling and resource optimization**. Strong background in **observability** (Prometheus, Grafana, ELK, OpenTelemetry), incident response, and driving platform reliability for high-availability systems. Skilled in leading SRE/DEVOPS teams, enforcing compliance standards, and building secure, multi-tenant systems.



Skills

- **Kubernetes & Cloud:** EKS, AKS, GKE, Helm, Cluster Management, Multi-region Deployments, Hub Spoke Architecture
- **Infrastructure as Code:** Terraform, Ansible .
- **Scripting:** Python, Bash .
- **Observability & Monitoring:** Prometheus, Grafana, ELK, Loki, OpenTelemetry, Datadog, Dynatrace
- **CI/CD & GitOps:** ArgoCD, Jenkins, Blue/Green & Canary deployments (Flagger)
- **Cloud Platforms:** AWS, Azure, GCP, Private Cloud (OpenStack)
- **Incident Management:** Root Cause Analysis, Self-healing systems, SLO/SLI design and enforcement



Work History

Jun 2024 - Current **Lead Site Reliability Engineer**
Grab Greco LLP, Bengaluru

Managing a multi-tenant Kubernetes platform powering Risk, Fraud Detection, Driver/Passenger Safety, and KYC systems, while leading B2B infrastructure for GrabDefence (defence.grab.com).

- **Architected and managed multi-region Kubernetes infrastructure on AWS EKS**, using Terraform for infrastructure-as-code and Datadog for comprehensive observability.
- **Built a GitOps platform with ArgoCD**, leading the migration of 100+ microservices from Jenkins to a declarative CI/CD workflow across regions.
- **Designed and executed synthetic stress tests with Locust**, simulating complete customer lifecycle flows (onboarding, transactions, fraud events)

to validate performance under peak load, including dynamic RDS scaling and failover testing.

- **Implemented fine-grained authorization using OpenFGA** in our **RAG-powered LLM system**, enabling secure, context-aware access control and delivering precise, tailored responses for GrabDefence end users.
- **Implemented Cilium for secure, scalable Kubernetes networking**, solving IP exhaustion issues with advanced CNI-based IP management and network policy enforcement.
- **Enabled progressive delivery using Istio + Flagger**, supporting automated canary deployments across high-risk services in a multi-tenant environment.
- **Designed and implemented BCDR (Business Continuity & Disaster Recovery) strategy for AWS EKS**, including cross-region failover, data replication, and automated recovery pipelines.
- **Built and managed GPU-based Kubernetes workloads with CUDA**, enabling ML model training at scale, with integrated **GPU observability using NVIDIA DCGM exporter + Prometheus** for real-time insights into utilization, memory, and thermals.
- **Drove cloud cost optimization** by implementing intelligent cluster autoscaling policies, VPA, and spot instance adoption across non-critical workloads.

Jun 2022 -
Jun 2024

Senior DevOps Engineer

Walmart Global Technology Services , Bengaluru, India

Key Contributions & Achievements: Team Leadership & Operations: Customer Success:

Responsible for building and managing the Walmart Cloud Native Platform (WCNP) to support containerized workloads across global eCommerce platforms, stores, distribution, and fulfillment centers — with Kubernetes as the core orchestrator.

- **Open Source Innovation:** Integrated Kubernetes SIG's Cluster Proportional Autoscaler into Walmart's platform, resulting in **\$10K/day cost savings** through optimized horizontal scaling.
- **Cluster Lifecycle Automation:** Automated node bootstrapping and cluster lifecycle management for **AKS and GKE clusters**, significantly improving scalability and reliability.
- **Developer Enablement:** Boosted developer productivity by delivering **self-service tooling (pipeline visualizer)** and optimizing CI/CD pipelines.
- **Security & Compliance:** Led **PCI compliance** initiatives for GKE environments, ensuring secure multi-tenant workloads aligned with industry standards.
- **Multi-cloud Optimization:** Designed and optimized Kubernetes deployments across **private cloud, Azure, and GCP**, achieving improved efficiency and cost control.
- **Observability & Policy Enforcement:** Deployed full-stack observability using **Prometheus, Grafana, and ELK**, and enforced security policies with **OPA and Kyverno**.
- **Disaster Recovery:** Implemented **Velero-based backup and recovery** solutions for Kubernetes CRDs with Azure Blob Storage, enhancing data resilience.
- **Platform Evangelism:** Conducted workshops and technical consultations to promote adoption of WCNP across internal teams.
- **Cluster Upgrades:** Led **Kubernetes version upgrades** for **5,000+ clusters on Azure**, ensuring zero downtime and minimal risk during rollouts.
- Acted as **Lead Consultant** for migrating legacy applications from VMs to **container-based microservices** in IDC.

Aug 2021 - **SDE2 (Devops Lead)**

May 2022

Maveric Systems Pvt Ltd., Bengaluru, India

- **Led cloud transformation initiatives** for banking clients by reducing on-premise footprint and enabling secure, scalable workloads on cloud-native platforms (AKS and GKE).
- **Built ephemeral Kubernetes-based testing environments** for developers, enabling on-demand, isolated environments for fast, reliable validation of code changes across AKS and GKE clusters. Provisioned using Terraform with full lifecycle automation and teardown to optimize resource usage.
- **Developed and supported container platforms** (Docker, AKS, GKE) to streamline application delivery across Windows, Linux, and containerized environments.
- **Built a PRP (Product Request Platform)** to automate delivery of developer-requested tools and products, packaged for Linux, Windows, and container use cases. Integrated **DevOps and SecOps workflows** to detect vulnerabilities prior to release, using automated scanning and approval gates.
- **Implemented DevSecOps best practices** for banking clients, embedding security controls across the CI/CD pipeline and enforcing compliance policies from code to production.
- **Designed and deployed a multi-environment Monitoring & Alerting system** for the Developer Platform, enabling granular visibility into deployment behaviors across DEV, STAGE, and PROD.
- **Established GKE security controls**, including workload identity, network policy enforcement, RBAC hardening, and CIS benchmark alignment to protect sensitive workloads in regulated environments.

Apr 2018 - **Senior Associate Consultant Devops**

Mar 2021

Infosys Pvt Ltd, Chandigarh, India

- **Provided L2/L3 incident support** for production environments, assisting the SRE team in root cause analysis of Kubernetes stack issues, node-level failures, and degraded cluster health.
- **Maintained and supported a suite of DevOps tools**, including Jenkins (**CloudBees & open-source**), **Git, Bitbucket, OpenShift, and container security scanning tools**, ensuring high availability and seamless development workflows.
- **Configured and managed CI/CD pipelines** using Jenkins and **CloudBees**, integrating **SCM systems such as Git, TFVC, Jazz, and build tools like Maven, Ant, and Make** across diverse project setups.
- **Scaled on-premise Jenkins Master/Agent architecture** using **OpenShift Container Platform**, optimizing resource utilization and enabling horizontal scaling with Kubernetes-native integrations and plugin automation.
- **Built auto-healing mechanisms by integrating AppDynamics** with Kubernetes clusters to monitor pod and application health, triggering automated remediation actions for infrastructure and application layer anomalies.
- Implemented best practices around configuration management, secure credentials handling, and SCM integration to support multi-team, multi-environment deployments at scale.

Oct 2015 - **Senior Associate Technology Devops**

Mar 2018

NAGARRO SOFTWARE PVT. LTD, GURGAON, INDIA

- **Developed and maintained Jenkins CI/CD pipelines** for continuous integration and deployment of microservices, integrating tools like JUnit for automated testing and SonarQube for code quality enforcement.

- **Configured Git with Jenkins**, using **POLL SCM** triggers to enable frequent code-based deployments, reducing manual intervention and speeding up release cycles.
- **Built Docker images** and custom Dockerfiles to package Java-based microservices, managing container lifecycle (snapshots, volumes, cleanup) for efficient runtime environments.
- **Automated environment provisioning** using **Ansible**, writing reusable playbooks in YAML to configure and deploy Dev/test infrastructure in a consistent and repeatable manner.
- **Supported 24/7 operations model**, actively participating in **non-business hours deployment automation** for web servers (Apache) using shell and Python scripts. These deployments directly supported **new feature rollouts**, with Python-based post-deployment checks to validate success and monitor revenue-impacting changes.
- Worked as part of **Airtel India's Feature Delivery Platform**, which enabled **add-on services** (e.g., value-added offers, content packs) for millions of users. Ensured fast and reliable delivery of features through robust CI/CD and infrastructure automation.
- Collaborated with Airtel engineering and NOC teams to conduct **failure simulations and high-availability testing** for datacenter-hosted services, validating HA configurations and failover readiness of critical applications.

Mar 2015 -
Oct 2015

VAS Engineer (Build and Release)

REALNETWORKS PRIVATE LIMITED, GURGAON, INDIA

- **Automated builds and deployments** with custom shell and Python scripts.
- **Managed versioning, tagging, and artifact promotion** in Git, SVN, and Artifactory.
- Collaborated with cross-functional teams to plan and execute smooth releases.
- Troubleshoot Jenkins job failures and improved pipeline stability.
- Integrated automated tests into CI pipelines for early defect detection.
- Documented processes to standardize builds and support team onboarding.
- Supported migration from Hudson to Jenkins to enhance CI scalability.

May 2011 -
Oct 2012

VAS Engineer Support (Linux Admin)

Comviva Technologies, Gurgaon, India

- **Maintained Linux-based infrastructure**, managing system services and ensuring availability of core DevOps tools such as Artifactory and Jenkins across production and staging environments.
- **Automated system tasks and backups** by writing robust scripts in **Python and Bash**, and scheduling jobs using **cron, SCP, SSH, and FTP** for secure and reliable data transfers and backups.
- **Managed JIRA instance administration**, including user/group permissions, workflow configuration, and custom field creation to support ITSM and development processes.
- **Installed and configured Apache Maven** on Linux systems and optimized pom.xml files for streamlined Java project builds as part of automated integration pipelines.
- **Automated Linux server provisioning and software deployments** using **Ansible**, improving consistency across environments and reducing manual configuration drift.
- Provided **day-to-day Linux system support**, including user and permission management, package updates, log analysis, and troubleshooting performance or service-related issues.



Certifications



Jun 2021 CKA: Certified Kubernetes Administrator



Jul 2020 AZ-900 Microsoft Certified: Azure Fundamentals



Education



Jun 2005 - Jun 2009 **B.Tech: Electronics And Instrumentation**
Kurukshetra University - Kurukshetra